

HILL VALLEY MULTI-ERA SANDBOX

Unified Master Technical Blueprint & Planning Elevation Document
Back to the Future × Vice City Hybrid — Web-First Architecture

Consolidated from 8 source documents · Generated 2026-08-18 17:53 · Version 1.0

1. Executive Vision & Design Pillars

This project creates a tightly-scoped, high-fidelity open-world playground centered on Hill Valley's Courthouse Square and its four immediate surrounding blocks. The core fantasy is instantaneous time travel between five canonical eras (1885, 1955, 1985, 2015, 2045) while driving a Vice City-style arcade vehicle. The technical foundation is deliberately web-first (PlayCanvas / React Three Fiber / pure WebGL + ES modules) to enable rapid AI-assisted iteration, mobile reach, and zero-install distribution. An optional Unreal Engine 5 path using World Partition + Data Layers is retained as a stretch goal for higher visual fidelity.

Non-Negotiable Design Pillars

- **Shared Geometry First** — One persistent master grid (roads, terrain heightmap, clock-tower structural shell, collision) remains loaded at all times. Only cosmetic skins, props, materials, lighting, and AI meshes change.
- **88 mph is the only load boundary** — Time jumps are pure asynchronous sub-level (or Data Layer) swaps concealed by a full-screen white flash + fire-trail particles. No traditional loading screens.
- **Arcade over Simulation** — Raycast suspension, low lateral friction under handbrake, direct velocity injection, and floaty bounce. Hover mode activates only in 2015/2045.
- **Causality is Data, not Narrative** — Butterfly effects are boolean flags (or bitmasks) that mutate asset visibility/state on subsequent era loads. No complex quest graph required for the core loop.
- **Scope is a feature** — 4x4 block radius maximum. Secondary streets and distant skyboxes are static or billboarded. Asset budget is controlled by modular naming and sequential loading.

2. Unified Project File Architecture (Single Source of Truth)

All prior document variants are collapsed into the following canonical tree. Every new subsystem must land inside this hierarchy.

hv-sandbox/	
index.html	# Shell + HUD markup + module bootstrap
css/	
time-circuits.css	# LED time circuits + keypad modal + flash overlay
js/	
main.js	# Optional thin orchestrator (or keep logic in index)
vehicle/	
RaycastArcadeController.js	# 4-pt raycast + drift + hover
temporal/	
TimeStateMachine.js	# 88 mph monitor + stream orchestration
ButterflyEffectManager.js	# Flag store + asset mutation rules
ParadoxEnforcementManager.js	# Anomaly score + stars + desaturation
ui/	
KeypadController.js	# Tab-triggered MMDDYYYY modal
audio/	
AudioManager.js	# Web Audio graph, Doppler, spatial lightning, radio
ai/	
TemporalPathfinder.js	# Era-aware node graph + pedestrian archetypes
ChasePhysicsController.js	# Predictive intercept + ram
loader/	
AssetPipelineLoader.js	# Sequential async queue + progress
storage/	
SaveStateCompressor.js	# Bitmask LocalStorage persistence

3. Core Data Model

3.1 Era Enumeration & Mapping

Canonical eras are strings: **1885 | 1955 | 1985 | 2015 | 2045**. Date entry (MMDDYYYY) is mapped by year ranges ($\leq 1920 \rightarrow 1885$, $1921-1970 \rightarrow 1955$, $1971-2000 \rightarrow 1985$, $2001-2030 \rightarrow 2015$, $>2030 \rightarrow 2045$). This mapping lives only inside `KeypadController.submitDate()` and is the single translation point from calendar input to asset layer.

3.2 Butterfly Flag Schema (Causality State)

Flags are stored in `ButterflyEffectManager.flags` and serialized via `SaveStateCompressor.bitfieldSchema`. Current canonical set (extensible):

- `pine_tree_destroyed_1885` $\rightarrow$ toggles Twin Pines / Lone Pine Mall signage in 1985+
- `courthouse_ledge_damaged_1955` $\rightarrow$ swaps pristine vs scarred ledge meshes from 1985 onward
- `diner_sign_vandalized_1985` $\rightarrow$ can drive 2015/2045 abandoned or corporate variants

3.3 Paradox / Wanted State

`ParadoxEnforcementManager` maintains: `anomalyScore` (float), `wantedLevelStars` (0–5), `paradoxTimer`, `maxParadoxDuration` (45 s). Stars are pure thresholds on `anomalyScore`. At 5 stars the desaturation shader ramps and a hard failure (reload) occurs if the timer expires.

4. Temporal Paradox Enforcement Logic — Deep Exploration

The paradox system is the primary gameplay loop after pure time-travel sightseeing. It converts anachronisms into a Vice City-style wanted escalation while remaining thematically faithful to the films’ “fading photograph” consequence.

4.1 Infraction Detection Matrix

Detection is continuous and frame-driven inside `ParadoxEnforcementManager.calculateTemporalInfractions(deltaTime)`. Current rules (easily extended):

Infraction	Condition	Score Rate	Design Intent
Hover in past	<code>currentEra < 2015 AND vehicle.hoverMode</code>	+25 / s	Most severe anachronism — flying car in 1955
Excessive speed in early eras	<code>currentEra < 1985 AND speedMPH > 50</code>	+8 / s	DeLorean is too fast for 1885/1955 traffic culture
Decay (no infraction)	No active infractions	−4 / s	Allows player to “cool down” by behaving

4.2 Star Thresholds & Escalation

`anomalyScore` thresholds: $10 \rightarrow 1\star$, $30 \rightarrow 2\star$, $60 \rightarrow 3\star$, $100 \rightarrow 4\star$, $150 \rightarrow 5\star$. At each change the HUD star string is rewritten ($\star$ repeated + $\blacksquare$ fillers). Visual and AI responses scale:

- **1 $\star$** — Pedestrians switch to panic/point animations (`TemporalPathfinder` archetype override).
- **2–3 $\star$** — Period-accurate enforcement vehicles spawn (sheriffs on horseback 1885, chrome sedans later). `ChasePhysicsController` activates.
- **4 $\star$** — Unmarked high-speed “temporal agents” with elevated `leadPredictionScalar`; they form perimeter behavior around Courthouse.
- **5 $\star$** — `paradoxWarningActive = true`; desaturation + mild blur shader ramps linearly with `paradoxTimer` / `maxParadoxDuration`. At 45 s $\rightarrow$ `triggerTimelineErasureFailureState()` (alert + `location.reload`).

4.3 Chase Physics (Intercept Algorithm)

`ChasePhysicsController` does not pure-pursue. It computes a lead point: `target = player.position + player.velocity  $\times$  (distance  $\times$  leadPredictionScalar)`. It then steers toward that point with high angular agility, clamps speed, and matches altitude when

the player is in hoverMode. On AABB proximity (< 4.2 units) it applies a velocity impulse to the player (rammingForceMultiplier) and self-separates to avoid sticky collisions. This produces the classic Vice City “they cut you off and smash” feel rather than endless tailing.

4.4 Identified Gaps & Recommended Elevations

- **Limited Infraction Vocabulary** — Only two continuous rules exist. Expand with discrete events: ramming period-correct vehicles, destroying era-key props, using future weapons/tools, or leaving a 2015 newspaper in 1955.
- **No “Fix the Paradox” Path** — Currently the only escape is to jump eras or wait for score decay. Add explicit repair actions (e.g., replace the pine tree, restore the ledge, return the stolen hoverboard) that instantly zero anomalyScore for that flag.
- **No Persistence of Wanted Across Jumps** — Stars reset on era change. Consider carrying a reduced “temporal heat” residue into the destination era so aggressive play still has consequences.
- **Shader is CSS-only** — Desaturation via canvas.style.filter works for the mock but should be replaced by a real post-process material / WebGL shader for production fidelity and performance.
- **Missing Multi-Agent Coordination** — Multiple chase vehicles currently act independently. A simple flocking or “box-in” coordinator would raise tension at 4–5★.

5. Investigation: Procedural Era Generation Algorithms

All source documents currently describe an **authored modular skinning** approach, not true procedural generation. The master grid is fixed; each era supplies pre-built façade variants, prop sets, material swaps, and lighting presets. Naming convention (HV_Block[X]_[Y]_[AssetType]_[Era]) enables deterministic lookup. This is correct for a high-fidelity, canonically accurate Hill Valley, but it leaves headroom for hybrid procedural techniques that reduce content authoring cost and increase replayability.

5.1 Current Authored Pipeline (Baseline)

- Persistent collision + road + sidewalk + clock-tower shell always loaded.
- Per-era layer nodes contain only deltas: materials, decals, props, signage meshes, vehicle/pedestrian meshes, particle emitters, lighting profiles.
- Butterfly flags simply enable/disable or swap named nodes at stream time.
- No runtime geometry generation; all meshes are pre-authored and streamed sequentially via AssetPipelineLoader.

5.2 Viable Procedural Layers (Recommended Hybrid)

True full-city procedural generation is out of scope and would destroy the “this is Hill Valley” feeling. Instead, apply procedural techniques only to secondary content that players notice but do not rely on for navigation or story:

- **Signage & Billboard Text** — Maintain a per-era lexicon + butterfly-aware templates. At load time, generate “Lone Pine Mall”, “Essex Thrift”, corporate 2045 names, etc., as texture atlases or 3D text meshes. Zero extra modeling cost.
- **Secondary Prop Density & Variation** — Use a simple Poisson-disk or grid-based scatterer driven by era parameters (trash density 1985 high, pristine 1955 low, drone pads 2015, wreckage 2045). Props are still pre-authored; only placement and count are procedural.
- **Pedestrian Appearance & Density** — TemporalPathfinder already returns movement archetypes. Extend with a weighted random mesh/skin selector per era + butterfly flags (e.g., more “survival” outfits if certain 2045 flags are set).
- **Weather & Atmospheric Intensity** — Parameterize acid-rain particle rate, fog density, dust amount, volumetric god-ray strength from a single era config object. Allows continuous variation and easy A/B testing.
- **Graffiti / Decal Sets** — Maintain small atlas libraries per era; at load time randomly (or deterministically from a seed derived from butterfly bitmask) choose which decals appear on which wall surfaces.
- **Radio & Ambient Audio Variants** — Already partially present; expand with procedural ad-lib commercial jingles or emergency broadcast snippets driven by the same flag state.

5.3 Explicitly Rejected for Scope Control

- Procedural road network generation (breaks the fixed Courthouse Square mental model).

- Procedural building massing / floor-plan generation (would lose recognizable BTTF landmarks).
- Full Wave Function Collapse or L-system cities (overkill and memory-hostile on mobile WebGL).

Recommendation: Keep the authored landmark + modular skin core. Add a thin ProceduralContentDirector that runs after ButterflyEffectManager.evaluateAndMutateAssets and before the white-flash fade-out. It receives the destination era string + current flag bitmask and mutates only secondary content. This preserves determinism for testing while dramatically reducing the number of unique assets that must be hand-authored.

6. Core Systems — Consolidated Specifications

6.1 Vehicle (RaycastArcadeController)

4-point (or averaged) raycast suspension using Hooke's law + damper. Forward/reverse forces injected directly into velocity. Handbrake drops lateralFriction to ~0.12 for frictionless slides. Angular turn is speed-scaled and direction-aware. Hover mode (2015/2045 only) replaces gravity with altitude spring targeting targetHoverHeight, adds air damping, and allows vertical thrust on combined forward+turn input. Speed is reported in MPH for the 88 mph trigger.

6.2 Time Engine (TimeStateMachine)

Monitors vehicle.speedMPH every tick. On ≥ 88 and destination $\neq$ current: lock input, store velocity, flash white, await streamLevelLayers (unload old era node, load new), enforce minimum 400 ms visual buffer, restore velocity with slight boost, fade flash, update HUD. streamLevelLayers is the single extension point for Butterfly evaluation, pathfinder rebuild, audio lightning, chase respawn, and save write.

6.3 UI / Input (KeypadController + CSS)

Tab toggles a modal numeric keypad. Digits append to an 8-char buffer; visual formatting inserts spaces. Enter validates month/day ranges, maps year $\rightarrow$ era, calls TimeStateMachine.setDestinationEra, updates the three LED rows. Clear and Backspace are supported. The CSS uses Share Tech Mono / Orbitron and authentic LED glow colors.

6.4 Audio (AudioManager)

Web Audio API graph: masterGain $\rightarrow$ destination. Engine is a sawtooth oscillator + lowpass, pitch-tracked to speed and hover state via setTargetAtTime. Spatial lightning is a procedural noise buffer through a panner (HRTF) and decaying gain. Listener is updated every frame from camera (or vehicle) transform. Multi-era radio stations are planned as separate AudioBufferSourceNodes switched by era.

6.5 Asset Pipeline & Save

AssetPipelineLoader enqueues {id, era, url}, processes sequentially with retries and 30 ms yields to keep the main thread responsive. Progress callback drives a full-screen loading UI. SaveStateCompressor packs the flag object into a bitmask, encodes to base-36, and stores a versioned JSON blob in LocalStorage. On boot the mask is restored before any era is loaded.

7. Refined Production Roadmap (Vertical-Slice Driven)

Previous roadmaps were phase-oriented (weeks 1–26). The elevated process replaces pure sequential phases with **vertical slices** that deliver a complete player experience as early as possible, gated by measurable criteria.

Slice 0 — Mechanical Core (Greybox) — Target: Week 2–3

- Persistent 4x4 grid + collision only (no era skins).
- RaycastArcadeController fully tunable; 88 mph triggers a console log + white flash only.
- Time circuits HUD shows static dates; Tab keypad writes destination era string.
- Success gate: drive continuously for 5 minutes at 60+ fps on mid-range mobile Safari; 50 consecutive 88 mph “jumps” (flash only) without crash.

Slice 1 — First Real Jump (1985 $\leftrightarrow$ 1955) — Target: Week 5–6

- Two full era layers (materials, key props, lighting).
- TimeStateMachine performs real node enable/disable + flash timing.
- One butterfly flag (pine tree) mutates the mall sign.
- Basic AudioManager engine loop + one lightning strike on arrival.
- Success gate: side-by-side visual comparison of 1955 vs 1985; jump feels instantaneous; memory stable after 30 jumps.

Slice 2 — Paradox Loop Live — Target: Week 8–9

- ParadoxEnforcementManager + star HUD + desaturation.
- ChasePhysicsController spawns and rams at 3★+.
- Hover mode functional in a temporary 2015 test layer.
- SaveStateCompressor persists flags across reloads.
- Success gate: player can intentionally raise to 5★, feel the chase, escape by jumping, and see the flag survive a browser refresh.

Slice 3 — Full Five Eras + Polish — Target: Week 14–16

- All five era skins + lighting + pedestrian archetypes.
- ProceduralContentDirector for secondary props/signage.
- Complete radio wheel + spatial ambient beds.
- Mobile touch controls + texture quality tiers.
- Success gate: full playable tour of all eras; 100 consecutive jumps; Lighthouse / WebGL performance budgets met.

Slice 4 — Content & Juice — Target: Week 20+

- Additional butterfly rules and “fix the paradox” interactions.
- Quest-lite loops (delivery, hover chase, photo opportunities).
- Final audio mix, particle polish, UI juice.
- Deployment to CDN + analytics for jump success rate and crash telemetry.

8. Evaluation of the Existing Planning Process

8.1 Strengths Observed Across the Eight Documents

- Extremely high thematic fidelity and clear player fantasy.
- Strong modular thinking (persistent grid + era skins) that correctly solves the “five open worlds” memory problem.
- Production-grade code snippets already exist for the hardest systems (raycast vehicle, time machine, paradox, chase, audio, sequential loader, bitmask save).
- AI-assistant-friendly structure (explicit file paths, mock objects, extension points).
- Scope discipline is repeatedly stated (4×4 blocks, no distant geometry).

8.2 Critical Weaknesses in the Planning Process Itself

- **Fragmentation** — Eight overlapping documents with evolving file trees, duplicate class names, and slight behavioral divergences (e.g., ButterflyEffectManager vs. TimelineStateFlags). No single source of truth existed until this document.
- **Phase-oriented rather than outcome-oriented** — Earlier roadmaps listed weeks of activity but lacked binary success gates, performance budgets, or “definition of done.”
- **Engine ambiguity** — Web-first code dominates, yet several documents still describe Unreal Data Layers and Niagara. Without an explicit Architecture Decision Record, contributors can waste effort on the wrong stack.
- **Missing risk & quality artifacts** — No performance budget table, no memory ceiling, no automated test plan for the 88 mph path, no legal/IP note (BTTF is heavily protected IP).
- **No vertical-slice discipline** — Features were specified in isolation. The first moment a player can feel “I just time-traveled and the world changed” was not forced early enough.

- **Procedural opportunity under-explored** — Authored content is correctly prioritized, but secondary procedural layers that would multiply content density at low cost were never systematically examined until Section 5 of this document.
- **Paradox system is thin** — The wanted-level loop is present but lacks depth (few infraction types, no repair path, no cross-era heat). This is the primary long-term engagement mechanic and needs deliberate design elevation.

9. Best Solutions to Elevate the Overall Planning Process

The single highest-leverage change is to treat this Unified Master Blueprint as the permanent single source of truth and to adopt a vertical-slice + gated process. Concrete elevations:

9.1 Immediate Process Changes (Do These First)

- **Lock the Architecture Decision** — Web-first is primary. UE5 is a documented stretch goal only. Any new work must target the JS/WebGL tree above.
- **Adopt this document as the living blueprint** — All future feature specs are written as appendices or PRs against this structure. Divergent documents are deprecated.
- **Introduce binary success gates** for every slice (see Section 7). No slice is “done” until the gate is measured and recorded.
- **Create a one-page Risk Register** covering: mobile WebGL memory, sequential load stalls, IP exposure, browser AudioContext autoplay policy, LocalStorage quota.
- **Define performance budgets** — e.g., initial payload < 40 MB compressed, steady-state < 180 MB heap on iPhone 12-class device, 60 fps while driving, < 16 ms for a full era swap.

9.2 Medium-Term Planning Elevations

- **Vertical-slice cadence** — Every 2–3 weeks a playable slice is demonstrated. Features that cannot be felt inside a slice are deferred.
- **Paradox design workshop** — Expand the infraction matrix and invent 3–5 concrete “fix the timeline” interactions before writing more chase code.
- **ProceduralContentDirector spike** — One week spike to prove signage + prop scatter driven by the butterfly bitmask. If successful, it becomes part of the stream pipeline.
- **Automated regression harness** — Headless or Playwright-driven tests that force 50 consecutive 88 mph jumps and assert no exceptions + stable memory.
- **Content naming & ownership matrix** — Explicit list of which blocks/landmarks are authored vs. procedurally varied, with owner and target polycount.

9.3 Long-Term Planning Maturity

- Add a lightweight Decision Log (ADR-style) for every non-obvious choice (why sequential loading, why bitmask save, why CSS flash vs. WebGL, etc.).
- Instrument telemetry from day one: jump success rate, average anomalyScore, most common 5★ trigger, crash signatures.
- Maintain a public (or internal) “Known Limitations” section so AI assistants and future collaborators do not re-propose rejected ideas (full procedural cities, five completely separate maps, etc.).
- Schedule a formal playtest after Slice 2 and again after Slice 3; feed qualitative findings back into the paradox and procedural systems before polish.

10. Conclusion & Immediate Next Action

The eight source documents already contain almost all of the technical material required to build a compelling, scope-controlled Hill Valley time-travel sandbox. What was missing was a single coherent blueprint, a clear primary engine decision, measurable gates, and deliberate attention to the two highest-leverage gameplay systems: temporal paradox enforcement depth and lightweight procedural content amplification.

This Unified Master Technical Blueprint now supplies that coherence. The recommended next action is not more documentation: it is to implement **Slice 0 (Mechanical Core)** against the file architecture in Section 2, measure the success gate, and only then expand into the first real era pair. Parallel to that implementation, a short design pass on the paradox repair loop and a one-week procedural signage/prop spike will raise the long-term engagement ceiling without expanding the core scope.

Primary stack: Web (PlayCanvas / R3F / pure WebGL) · Persistent 4×4 grid · 5 authored era skins · Butterfly bitmask causality · Paradox wanted loop · Sequential asset pipeline · LocalStorage persistence.

— End of Unified Master Technical Blueprint v1.0 —